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A

Let A = cross-sectional area of piston

Since no change in temperature,

$$p_1 V_1 = p_2 V_2$$

$$p_2 = (100 \text{ kPa} \times 80 \text{ mm} \times A) / (160 \text{ mm} \times A) = 50 \text{ kPa}$$

$$\begin{aligned}\therefore F &= (\text{atm pressure} - p_2)A \\ &= (100 - 50) 10^3 \times 3 \times 10^{-3} \\ &= 150 \text{ N}\end{aligned}$$

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A